

NCIE-011 Skeletal v0.1 → Full v1.0

Expansion Verification Report

API, Integration & Data Connector Specification — IN DEVELOPMENT / FOR HUMAN REVIEW OF FULL EXPANSION

1. Scope Verification

NCIE-011 v1.0 is a controlled full expansion of the approved NCIE-011 Skeletal v0.1: every approved chapter purpose, Salient Things to Note, principal component, upstream dependency/authority and Human Review focus is elaborated into API, event, connector and integration-contract architecture, without inventing gateway, broker, schema-registry, API-management, connector, ETL, webhook or service-mesh products, without hard-coding REST/GraphQL/gRPC/Kafka/AMQP or other protocols as universal requirements, and without inventing rate limits, quotas, timeouts, retry counts, backoff periods, payload sizes, retention windows, deprecation periods, buffering limits or SLOs.

2. 32-Chapter Integrity Verification

A programmatic scan of the final document confirms exactly 32 sequential 'NCIE-011 — Chapter N Expansion' labels, N = 1..32, each present exactly once, in the approved order, with no chapter added, removed, merged, split or renamed relative to the approved Skeletal v0.1 structure.

3. Salient Things to Note Hard-Gate Verification

Per directive item 75 — continuing the discipline established after the NCIE-009 Chapter-27 omission — every chapter must retain an explicit '<N>.0 Salient Things to Note' subsection. This build enforced the requirement structurally: the chapter-rendering helper (lib.py's add_chapter function) raises a build-time assertion error if a chapter's salient content is omitted, so the document cannot be assembled with a missing subsection. A programmatic scan of the final assembled document, rerun independently rather than inferred from the build script succeeding, confirms an explicit Salient Things to Note heading present for all 32 chapters (N = 1 through 32), with zero missing.

4. Skeleton Fidelity Verification

Each of the 32 chapters was authored directly from its approved skeletal Chapter Purpose, Salient Things to Note, Principal Components, Upstream Dependencies/Authority and Human Review Focus (see the accompanying Change Register for the per-chapter mapping). No approved chapter concept was omitted, and none was conceptually inverted.

5. No Product, Protocol or Threshold Invention

A full-text scan for named gateway, broker, schema-registry, API-management, connector, ETL, webhook or service-mesh products found none invented (directive item 8). A separate scan for REST/GraphQL/gRPC/Kafka/AMQP or other protocols hard-coded as universal requirements found none — every chapter states contract properties (synchronous/asynchronous mode, delivery semantics, ordering) rather than a mandated protocol (directive item 9). A third scan for rate limits, quotas, timeouts, retry counts, backoff periods, payload sizes, retention windows, deprecation periods, buffering limits or SLOs found none invented (directive item 10) — every such matter is instead recorded as a named Human Review item (e.g., HR11-17-1 for transfer thresholds, HR11-20-1 for deprecation windows, HR11-26-1 for quota ownership, HR11-29-1 for buffering tolerances).

6. Governing Integration Invariants Verification

The 20 invariants proposed in the approved Skeletal v0.1 ('Proposed Governing Invariants') are restated unchanged in front matter §0.3, and are additionally cross-referenced — not restated — against the 12 NCIE-008 governance invariants, the 16 NCIE-009 security invariants, the 19 NCIE-010 Evidence/provenance/audit invariants, and other upstream invariants already binding from NCIE-005/006/007. A full-text scan confirms every invariant is both stated once in §0.3 and applied by name in at least one operative chapter (e.g., API ACCESS ≠ AUTHORIZATION in Ch.27; REQUEST ACCEPTED ≠ ACTION COMPLETED in Ch.12, 23; EVENT RECEIVED ≠ AUTHORITATIVE CURRENT STATE in Ch.18; SCHEMA COMPATIBILITY ≠ SEMANTIC COMPATIBILITY in Ch.20; RETRY ≠ SAFE REEXECUTION in Ch.22).

7. Integration/Upstream Boundary Verification

Chapter 1 Table 1 fixes what NCIE-011 does not own (canonical data, business-domain logic, ARGUS behavior, Memory semantics, Agent/VPF semantics, institutional governance, security enforcement, Evidence meaning, UX, storage and operations — all NCIE-001–010's). Every consequential interface control in Chapters 4–30 is expressed as enforcement of a named NCIE-008 decision-right, NCIE-009 security control, or NCIE-010 Evidence rule with the accountable holder marked 'to be assigned,' never as NCIE-011 assigning new institutional authority, security mechanisms or Evidence semantics of its own (directive item 7).

8. Outcome-Unknown Audit

A full-text scan (directive item 63) for wording that would treat accepted/submitted/timeout responses as confirmed Tool/external outcomes found none — Chapter 12 §12.2, Chapter 23 Table 28 and Chapter 25 Table 31 each preserve OUTCOME UNKNOWN as an explicit, non-collapsible state requiring reconciliation, never silently resolved to success.

9. Event-Authority Audit

A full-text scan (directive item 64) for wording that would make an event authoritative current state without reconciliation found none — Chapter 18 §18.1 and Table 22's reconciliation column require every event category to state what a consumer must reconcile against, consistent with EVENT RECEIVED ≠ AUTHORITATIVE CURRENT STATE.

10. Retry-Safety Audit

Verified (directive item 65) that retry/replay cannot silently duplicate consequential side effects or reuse stale approvals/authorization: Chapter 22 confirms RETRY ≠ SAFE REEXECUTION and IDEMPOTENCY KEY ≠ BUSINESS UNIQUENESS, and Chapter 12's Tool execution receipts require reconciliation before an outcome is treated as final, preventing a blind retry from reusing a stale approval reference.

11. Schema-Semantic Audit

Verified (directive item 66) that schema compatibility is never presented as semantic/business compatibility: Chapter 20 §20.1 explicitly states SCHEMA COMPATIBILITY ≠ SEMANTIC COMPATIBILITY, and confirms contract versions (API VERSIONING ≠ DATA-MODEL VERSIONING) never redefine canonical entity semantics owned by NCIE-003.

12. Security Audit

Verified (directive item 67) that all API/event/connector surfaces preserve NCIE-009 current authn/authz, DLP, scoped identities, secrets handling, revocation and protected-data controls: Chapter 27 Table 34 (API Authorization Matrix) binds every caller principal class defined in this document to a current-authorization check and an audit requirement, with no principal class exempted by default.

13. Evidence/Audit Interface Audit

Verified (directive item 68) that NCIE-010 Evidence/custody/audit interfaces preserve Evidence identity, chain-of-custody, historical integrity and provenance/audit distinctions: Chapter 14 confirms API design cannot redefine Evidence semantics and that every operation maps one-to-one to an NCIE-010 lifecycle event rather than introducing a new one.

14. Interface Catalogue and Service Contract Matrix Verification

Chapter 3 Table 4 (Interface Catalogue Matrix, directive item 53) catalogues 16 interface classes across every chapter of this document with owning service, caller/consumer class, mode, schema reference and authorization boundary. Table 5 (Service Contract Matrix, directive item 54) maps 5 contract classes to owning service and schema references without NCIE-011 becoming the canonical data owner.

15. Connector, Event Contract, Async-State and Error-State Matrix Verification

Chapter 15 Table 18 (Connector Matrix, directive item 57) maps 4 connector classes to identity, scope, provenance and security without inventing institutional actors. Chapter 18 Table 22 (Event Contract Matrix, directive item 56) maps 8 event categories to publisher, delivery semantics and reconciliation requirement. Chapter 23 Table 28 (Asynchronous-State Matrix, directive item 59) fixes 7 consistent long-running-operation states and their valid transitions. Chapter 25 Table 31 (Error-State Matrix, directive item 58) fixes 9 error classes with retryability and caller-safe message discipline, none exposing sensitive internal detail by default.

16. Test-Status Integrity Verification

Chapter 30 Table 38 specifies 8 test classes (contract, consumer compatibility, authorization, event replay, duplicate suppression, retry safety, connector fault injection, external sandbox), each traced to the chapter(s) it verifies. Every test is marked TEST SPECIFIED only; a full-text scan confirms no test anywhere in this document is marked TEST EXECUTED, TEST PASSED, or claims IMPLEMENTATION EVIDENCE AVAILABLE (directive items 62 and 71) — detailed test-case authoring, execution and partner certification are explicitly deferred to NCIE-016.

17. No Fabricated Protocols, Products or Implementation Claims

A full-text scan confirms no API, connector or event surface anywhere in this document is claimed implemented, partner-certified, operational or production-ready merely because this v1.0 specifies it (directive item 71) — every RACI/control-ownership table marks its holder 'to be assigned,' and Chapter 30 explicitly marks every test SPECIFIED, not executed.

18. Human Review Register Mechanical Reconciliation (Correct From First Build)

Chapter 31's register was recomputed programmatically by scanning the Human Review Focus bullet text already rendered in Chapters 1-30, plus HR11-31-1 (Chapter 31's own item) and HR11-32-1 (Chapter 32's own item) — both defined once and included in the register-construction list from the start of the build, a design choice made specifically to avoid

repeating the register-reconciliation defect found and corrected in the companion NCIE-008 v1.0 document's initial build. Result: 32 total items, 24 Blocking / 8 Non-Blocking, 0 duplicate Review IDs, 0 orphan Human Review items. Classification breakdown: 19 Institutional Confirmation Required, 5 Security/Sovereignty Confirmation Required (Chapters 4, 21, 24, 25, 27), 1 Legal/Policy Confirmation Required (Chapter 7 operator classes), 6 Engineering/Infrastructure Discovery Required (Chapters 16, 17, 20, 26, 28, 29), 1 Resolved by Upstream Baseline (Chapter 32).

19. Upstream Human Review Deduplication (Directive Item 61)

HR11-1-1 (confirm NCIE-011 acceptance and NCIE-013 overlap resolution) restates the identical institutional question NCIE-008 already opened as HR8-10-1 (acceptance authorities for each NCIE-series document and chapter-level sign-off); Chapter 31 Table 41 cross-references HR8-10-1 rather than treating HR11-1-1 as an independent open question. Every other NCIE-011 Human Review item was individually checked against the NCIE-008/009 Human Review Registers and found to concern an interface/integration-specific decision that NCIE-008/009 do not themselves resolve — no other duplicate was found.

20. Blocking Scope Discipline Verification

Each Blocking item in the Chapter 31 register states precisely which threshold, ownership assignment, security control or acceptance authority it blocks, rather than implying the whole document or the whole approved upstream chain is blocked. Non-Blocking items (Chapters 2, 8, 11, 16, 19, 23, 28, and the resolved Chapter 32 item) are explicitly marked as not gating the corresponding chapter's integration operation in the interim.

21. Traceability and Downstream Non-Preemption Verification

Chapter 32 Table 42 traces all 32 chapters to their primary upstream authority (approved NCIE-001-010 and/or the approved NCIE-011 Skeletal v0.1); no chapter states an interface/integration requirement without an identifiable source. Table 43 confirms this document hands over UX-facing contracts to NCIE-012, domain connector patterns to NCIE-013, logical retention interfaces to NCIE-014, operational responsibilities to NCIE-015, and the 8 API/integration test classes to NCIE-016 — without specifying their technical implementation, consistent with directive item 20 (downstream non-preemption).

22. Version, Document-Control and Running-Header Verification

Cover and Document Governance table confirmed reading: Edition 'In development — Version 1.0 (Full Expansion of Approved Skeletal v0.1)'; Requirements/Architecture/Canonical-data/Engineering/ARGUS-behavioral/Memory-Context/Orchestration/Governance/Security/Evidence-provenance-audit authority rows all read their approved current versions (NCIE-001 v1.3 through NCIE-010 v1.0); a 'Prior baseline' row records the approved Skeletal v0.1 as historically preserved; Status reads IN DEVELOPMENT — FOR HUMAN REVIEW OF FULL EXPANSION. The running header — rendered from a copied NCIE-010 v1.0 template to inherit this series' visual styling, including the logo, centered cover text, page break before Document Governance, and non-splitting governance-table rows already corrected in that series — was corrected to 'NCIE-011 | API, INTEGRATION & DATA CONNECTOR SPECIFICATION | VERSION 1.0' and the footer corrected from 'CONTROLLED EVIDENCE DOCUMENT' to 'CONTROLLED INTEGRATION DOCUMENT' and from 'NCIE-010' to 'NCIE-011' before this document was rendered to PDF.

23. Full-Document Cross-Reference Scan

Every occurrence of 'NCIE-010' in the final document was individually inspected (47 occurrences outside Upstream/Trace blocks). Each was confirmed to be a legitimate Evidence/provenance/audit citation (e.g., 'NCIE-010 Ch.3/7/19', 'NCIE-010 acquisition provenance', 'Reconcile against NCIE-010 Evidence state') consistent with directive item 7's requirement that NCIE-011 consume, not redefine, NCIE-010 Evidence semantics — none is a leftover donor-template branding error. The cover, header, footer and Document Governance 'Document' row were separately confirmed to read NCIE-011, not NCIE-010.

24. Character-Encoding Integrity Check

Following the same build pipeline used for the whole NCIE series (UTF-8 source fragments executed via a Python exec() call with an explicit encoding='utf-8' argument, a fix first required during the NCIE-008 build), a full-document scan of the final NCIE-011 document confirmed zero mojibake occurrences across all 646 paragraphs and 44 tables.

25. Visual and Structural Verification

The complete document was opened in Microsoft Word, its Table of Contents field (a live Word field, not static text) was updated and confirmed to populate with correct chapter/section titles and real page numbers, repaginated to 58 pages, saved, and rendered to PDF. Independent text extraction from the rendered PDF confirms the cover occupies page 1 alone and Document Governance begins on page 2 (the manual page break), with the running header and footer correctly reading NCIE-011 branding from page 2 onward. No table clipping, overlapping text, or broken table was observed across the seven consolidated matrices, the Chapter 31 Human Review Register tables, or the Chapter 32 traceability/handover tables.

26. Final Human Review Extraction (Authoritative)

Performed directly against the final specification file by independently scanning every paragraph and table cell for the pattern HR11-<chapter>-<n>: TOTAL IDS FOUND = 32. UNIQUE IDS = 32. DUPLICATES = 0. OPEN = 31. RESOLVED = 1 (HR11-32-1, Resolved by Upstream Baseline). BLOCKING = 24. NON-BLOCKING = 8. CLASSIFICATION TOTALS: Institutional Confirmation Required = 19; Security/Sovereignty Confirmation Required = 5; Legal/Policy Confirmation Required = 1; Engineering/Infrastructure Discovery Required = 6; Resolved by Upstream Baseline = 1. ORPHAN IDS = 0 — every one of the 32 unique IDs found anywhere in the document also appears as a row in the Chapter 31 Complete Register table, which itself contains exactly 32 rows with no duplicate Review ID.

27. Three-Artifact Consistency Check

The NCIE-011 v1.0 main document, this Verification Report and the accompanying Full-Expansion Change Register agree exactly on: Source Version v0.1 (Skeletal) → Target Version v1.0 (Full); Authority chain approved NCIE-001 v1.3 through approved NCIE-010 v1.0, plus the approved NCIE-011 Skeletal v0.1; Status IN DEVELOPMENT — FOR HUMAN REVIEW OF FULL EXPANSION (not Approved); 32 chapters preserved in order (0 missing, 0 duplicate, 0 added, 0 unauthorized renaming); Salient Things to Note hard gate = 32/32 explicit subsections present; Human Review Register TOTAL = 32, BLOCKING = 24, NON-BLOCKING = 8, classification totals as in §26 above; orphan Human Review IDs = 0; duplicate Human Review IDs = 0; API/integration test-class status = 8 tests specified, none executed or passed; integration invariants = 20 NCIE-011-specific plus cross-referenced upstream invariants; downstream handover = NCIE-012/013/014/015/016 each receive a defined interface without technical preemption. No mismatch was found between the three artifacts.

28. Final Readiness Disposition

NCIE-011 — API, Integration & Data Connector Specification v1.0 — Full Expansion of Approved Skeletal v0.1 — is READY FOR HUMAN REVIEW. The approved v0.1 skeleton remains historically preserved as the expansion architecture; every proposed threshold, product/protocol choice and ownership assignment deferred in Chapters 1-30 remains a proposal pending resolution of the Chapter 31 Human Review Register; every §30.1 test remains SPECIFIED, not executed, pending NCIE-016. This document is NOT marked Approved.